Curriculum Vitae

Ayato Kitadai

Last updated: 2026/7/1

Education

- PhD Program, Department of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo, Apr. 2025 - Present
- Master's Program, Department of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo, Apr. 2023 - Mar. 2025
- Bachelor's Program, Department of Systems Innovation, Faculty of Engineering, The University of Tokyo, Apr. 2019 - Mar. 2023

Fellowships and Support

- JSPS Research Fellowship for Young Scientists (DC1), Apr. 2025 - Present
- WINGS GLAFS Excellent Research Assistant, Sep. 2023 - Mar. 2025

Qualifications

- Teaching certificate, junior and senior high school mathematics

Research Areas

Service Engineering Mechanism Design

Game Theory Market Design

Experimental Economics

Large Language Models

Multi-Agent Simulation

Peer-Reviewed Publications (15)

1. **A. Kitadai**, S.D.R. Lugo, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Can AI with High Reasoning Ability Replicate Human-like Decision Making in Economic Experiments?” *Group Decision and Negotiation*, 34(6), 1303-1326, 2025.
2. H. Sawazaki, X. Shang, S. Lee, **A. Kitadai** and N. Nishino; “Tenant allocation in shopping malls based on deferred acceptance algorithm,” *Procedia CIRP*, 138, 3-8, 2026.
3. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; “Profitability of Slow Steaming Under Vessel Offshore Waiting Conditions,” *IEEE Big Data*, pp. 5947-5953, 2025.
4. T. Nakashima, T. Oyama, R. Wada, **A. Kitadai** and N. Nishino; “A Transdisciplinary Approach to Cross-Sectoral Collaboration for Climate Change Adaptation: Bridging Synergies and Trade-Offs,” *Advances in Transdisciplinary Engineering*, 76, 428-437, 2025.
5. **A. Kitadai**, S.D.R. Lugo and N. Nishino; “Matching Problems in Remanufactured Products for Product Service Systems in the Circular Economy,” *IFIP Advances in ICT*, 767, 246-257, 2025.
6. S. Lee, **A. Kitadai**, Ye Cheng and N. Nishino; “Transformation of Many-to-One to One-to-One Matching Problems Through Replication,” *IFIP Advances in ICT*, 767, 287-301, 2025.

7. **A. Kitadai**, K. Ogawa and N. Nishino; “Examining the Feasibility of Large Language Models as Survey Respondents,” *IEEE Big Data*, pp. 3858-3864, 2024.
8. Y. Takenoya, **A. Kitadai** and N. Nishino; “Accounting for Traffic Congestion: Heterogeneous Target Arrival Times, Uncertainty, and Monetary Incentives,” *IEEE Big Data*, pp. 3880-3888, 2024.
9. M. Kobayashi, **A. Kitadai**, U. Sato, M. Fujita and N. Nishino; “Real Traffic Analysis Around Large-Scale Shopping Mall Using Game Theory,” *IFIP Advances in ICT*, 730, 408-423, 2024.
10. M. Fujita, Z. Cheng, S.D.R. Lugo, **A. Kitadai**, Y. Dai, Y. Fukasawa and N. Nishino; “Analysis of Medical Data and Machine-Learning Algorithms from the Perspective of Public-Goods Models of Data-Provision Decision Making,” *Procedia CIRP*, 126, 3-8, 2024.
11. M. Fujita, Y. Dai, **A. Kitadai**, S.D.R. Lugo, Z. Cheng and N. Nishino; “Survey on Stakeholder Cooperative Behavior for Designing Voluntary Medical Data Provision Motivation Mechanisms,” *Procedia CIRP*, 126, 14-19, 2024.
12. **A. Kitadai**, S.D.R. Lugo, S. Lee, M. Fujita and N. Nishino; “Weighted Reward Allocation Mechanism for Data Collection Quality,” *Lecture Notes in Business Information Processing (LNBIP)*, 478, 161-174, 2023.
13. **A. Kitadai**, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Toward a Novel Methodology in Economic Experiments: Simulation of the Ultimatum Game with Large Language Models,” *IEEE Big Data*, pp. 3168-3175, 2023.
14. Z. Zhou, M. Kobayashi, U. Sato, K. Akashi, **A. Kitadai**, S. Sugihara, Y. Fukasawa, M. Fujita and N. Nishino; “Evaluating the alleviation of traffic congestion through Bus Rapid Transit using Multi-Agent Simulation,” *IEEE Big Data*, pp. 3237-3244, 2023.
15. M. Fujita, **A. Kitadai**, K. Sumikura and N. Nishino; “Toward the Utilization of Artificial Intelligence in Life Sciences: A Voluntary Medical Data Provisioning Model,” *IEEE Big Data*, pp. 3250-3257, 2022.

Working Papers (7)

1. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” arXiv:2606.03030, 2026.
2. **A. Kitadai**, S. Yoshimura, T. Nakashima, N. Torpo, R. Miratsu, N. Mizutani and N. Nishino; “‘Sail Fast, Then Wait’ in First-come, First-served Port Queues: Information Sharing for Sustainable Shipping,” arXiv:2602.01958, 2026.
3. **A. Kitadai**, T. Ito, Y. Nagoh, H. Takahashi, M. Fujita, S. Lee, F. Miyahara, T. Natsume and N. Nishino; “Decision Support System for Technology Opportunity Discovery: An Application of the Schwartz Theory of Basic Values,” arXiv:2602.11855, 2026. R&R at *Technology in Society*.
4. T. Nakashima and **A. Kitadai**; “Adaptation Beliefs and Defensive Over-Adaptation under Nonlinear Climate Risk.” Under review.
5. **A. Kitadai**, Y. Fukasawa and N. Nishino; “Do Rich Persona Data Improve the Calibration of LLM Agents to Human Behavior? Evidence from the Ultimatum Game.” Under review.
6. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?” Under review.
7. **A. Kitadai** and Y. Nakamoto; “The Effect of AI-Mediated Institutional Communication on Decision-Making in a Dynamic Common Pool Resource Game.” Under review.

Conference Presentations (International, Oral, Peer-Reviewed) (9)

1. **A. Kitadai**, S. Yoshimura, T. Nakashima, N. Torpo, R. Miratsu, N. Mizutani and N. Nishino; "'Sail Fast, Then Wait' in First-come, First-served Port Queues: Information Sharing for Sustainable Shipping," GDN 2026, Katowice, Poland, June 2026.
2. J. Zhang, **A. Kitadai** and N. Nishino; "The Dynamics of Coopetition: A Biform Game Analysis of Incentive Mechanisms in DevOps," GDN 2026, Katowice, Poland, June 2026.
3. **A. Kitadai** et al.; "Information Asymmetry in FCFS Port Queues: A Game Theoretic Analysis of 'Sail Fast, Then Wait'," SMRC x MTEC/ICMASS 2026, Singapore, April 2026.
4. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; "Analysis of Optimal Ship Navigation in Offshore Waiting Situations," ICServ 2025, Stockholm, December 2025.
5. H. Cheng, S. Lee, H. Watanabe, **A. Kitadai**, Y. Nagoh, M. Fujita and N. Nishino; "How Can Barriers to Innovation Diffusion and Transition Be Identified from Bibliometric Data?," R&D Management Conference 2025, July 2025.
6. **A. Kitadai**, T. Nakashima, Y. Tsurusaki, R. Miratsu, N. Mizutani and N. Nishino; "Can rich information decrease anchorage waiting time? A game-theoretic analysis toward sustainable shipping," EURO 2025, Leeds, UK, June 2025.
7. **A. Kitadai**, R. Ishikawa, S. Lee, J. Teng, K. Nishiyama and N. Nishino; "Consensus Formation in Physical, Virtual, and Interverse Spaces: Experimental Insights," GDN 2025, Zaragoza, Spain, June 2025.
8. H. Sawazaki, X. Shang, S. Lee, **A. Kitadai** and N. Nishino; "Tenant allocation in shopping malls based on deferred acceptance algorithm," 18th CIRP ICME, Naples, Italy, July 2024.
9. **A. Kitadai**, S.D.R. Lugo, Y. Tsurusaki, Y. Fukasawa and N. Nishino; "Can AI with High Reasoning Ability Replicate Human-like Decision Making in Economic Experiments?," GDN 2024, Porto, Portugal, June 2024.

Conference Presentations (Domestic, Oral) (25)

1. **A. Kitadai**, Y. Kumagai and R. Yasunaga; "Bounded Rationality in Large Language Models: Price-Quality Heuristics and Self-Control," JSAI 2026: The 40th Annual Conference of the Japanese Society for Artificial Intelligence, Gunma and online, June 2026.
2. Y. Hoshino, **A. Kitadai** and N. Nishino; "Service Design for Proxy Negotiation Markets with AI Agents: Validating the Effectiveness of Matching Mechanisms," JIMA Spring Conference 2026, Hakodate, June 2026.
3. Y. Takenoya, **A. Kitadai** and N. Nishino; "A Screening Model of Upfront Contracts for Agents Who Ignore Convex Inequalities," JIMA Spring Conference 2026, Hakodate, June 2026.
4. **A. Kitadai** and N. Nishino; "The Effect of Introducing a Free-Dialogue Chatbot for Supporting Understanding of Matching Mechanisms on People's Decision-Making," JIMA Spring Conference 2026, Hakodate, June 2026.
5. **A. Kitadai**, Y. Nakamoto and N. Nishino; "Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?," 29th Experimental Social Science Conference, Kochi, May 2026.
6. Y. Hoshino, **A. Kitadai** and N. Nishino; "Service Design for Proxy Negotiation Markets with AI Agents: Validating the Effectiveness of Matching Algorithms," 14th Annual Conference of the Service Science Society of Japan, Mar. 2026.

7. Y. Yoshida, **A. Kitadai** and N. Nishino; “Game-Theoretic Analysis of Dynamic Production Planning and Core Collection Strategies in Commercial Vehicle Remanufacturing,” 14th Annual Conference of the Service Science Society of Japan, Mar. 2026.
8. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; “Profitability of Slow Steaming Under Vessel Offshore Waiting Conditions,” 14th Annual Conference of the Service Science Society of Japan, Mar. 2026.
9. Y. Yoshida, **A. Kitadai** and N. Nishino; “Environmental Impact of Remanufacturing in the Commercial Truck Industry,” JIMA Fall Conference 2025, Oct. 2025.
10. **A. Kitadai**, R. Ishikawa, S. Lee, J. Teng, K. Nishiyama and N. Nishino; “New Perspectives on Service Design for the Virtual Economy,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 2025.
11. H. Watanabe, **A. Kitadai**, Y. Nagoh, S. Lee, T. Natsume, F. Miyahara, M. Fujita and N. Nishino; “A Decision Support Tool for New Product Development Using LLMs and Matching Models,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 2025.
12. Z. Zhou, **A. Kitadai**, U. Sato, Y. Takenoya, M. Fujita and N. Nishino; “Evaluation of Bus Rapid Transit Introduction for Traffic Congestion Mitigation,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 2025.
13. Y. Takenoya, **A. Kitadai** and N. Nishino; “Analysis of Monetary Incentives for Congestion Mitigation in a Departure-Time Choice Model,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 2025.
14. T. Nakamura, **A. Kitadai** and N. Nishino; “Theoretical Analysis and Economic Experiment on Cooperative Behavior in Data Collection: A Nonlinear Public Goods Game Under Incomplete Information,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 2025.
15. **A. Kitadai**, Y. Fukasawa and N. Nishino; “Simulation of the Ultimatum Game with Large Language Models: Reproducibility of Experimental Results Through Bias Adjustment,” 18th Annual Meeting of the Behavioral Economics Society, Osaka, Dec. 2024.
16. **A. Kitadai** and N. Nishino; “Substitutability of Large Language Models as Survey Respondents,” JIMA Fall Conference, Shimane, Oct. 2024.
17. Y. Takenoya, **A. Kitadai** and N. Nishino; “Numerical Analysis of Traffic Evaluation Incorporating Heterogeneous Target Arrival Times,” JIMA Fall Conference, Shimane, Oct. 2024.
18. Z. Zhou, **A. Kitadai**, U. Sato, Y. Takenoya and N. Nishino; “Evaluation of Effective Bus Rapid Transit Introduction for Traffic Congestion Mitigation,” JIMA Fall Conference, Shimane, Oct. 2024.
19. M. Kobayashi, **A. Kitadai**, Z. Zhou, M. Fujita and N. Nishino; “Traffic Analysis for Congestion Mitigation Around Large-Scale Shopping Facilities,” 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 2024.
20. Z. Cheng, **A. Kitadai**, S.D.R. Lugo, N. Nishino, M. Fujita and L. Zhang; “Medical Data Classification and Provision Willingness Analysis Based on Public Goods Perspective,” 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 2024.
21. Y. Cheng, H. Sawazaki, N. Nishino, **A. Kitadai** and S. Lee; “Transformation of Many-to-One Matching Problems into One-to-One Matching Problems Through Replication,” 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 2024.

22. **A. Kitadai**, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Simulation of the Ultimatum Game with Large Language Models: Toward a New Methodology in Economic Experiments,” 17th Annual Meeting of the Behavioral Economics Society, Kochi, Dec. 2023.
23. **A. Kitadai**, M. Fujita and N. Nishino; “Designing a Mechanism for Smooth Data Collection,” 11th Annual Conference of the Service Science Society of Japan, Kyoto, Mar. 2023.
24. T. Ito, N. Nishino, H. Takahashi, **A. Kitadai**, M. Fujita, Y. Nagoh, T. Natsume and F. Miyahara; “How Do Technologies Create Value? A Function-Value Matching Model and Case Study,” 11th Annual Conference of the Service Science Society of Japan, Kyoto, Mar. 2023.
25. M. Fujita, **A. Kitadai**, K. Sumikura and N. Nishino; “Designing Mechanisms to Facilitate Medical Data Provision for AI Utilization in Life Sciences,” 21st SIG-BI, Japanese Society for Artificial Intelligence, Kanagawa, Sep. 2022.

Awards (7)

1. 2026 GDN Springer Best Student Paper Award (2026)
2. Head of Department Award (Academic Achievement), Dept. of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo (2025)
3. 2024 GDN Springer Best Student Paper Award (2024)
4. Best Paper Award and Arai Award, 12th Annual Conference of the Service Science Society of Japan (2024)
5. Poster Presentation Encouragement Award (General Division), Behavioral Economics Society (2024)
6. 2023 GDN Springer Best Student Paper Award (2023)
7. Best Paper Award, 11th Annual Conference of the Service Science Society of Japan (2023)

Talks and Other Activities (12)

1. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?,” International Workshop on Human-AI Interaction, ISER, Osaka, March 24, 2026 (invited).
2. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” International Workshop on Human-AI Interaction, ISER, Osaka, March 23, 2026.
3. **Ayato Kitadai**, “Exploring Various Methods for Generative Agents to Replicate Results in Economic Experiments of the Ultimatum Game,” HSE Comsoc seminar, DeCAn, February 12, 2026 (invited).
4. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?,” Kansai University RISS Workshop, February 28, 2026.
5. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” Kansai University RISS Workshop, February 28, 2026.
6. **Ayato Kitadai**, “Substitutability of Human Behavior by LLM Agents: When and When Not to Use Silicon Sampling,” invited seminar, Analytical Marketing Research Division, Japan Marketing Science Association, October 7, 2025.
7. **Ayato Kitadai**, “AI-driven ‘Human-like’ Decision Making: Utilizing Insights and Data from Behavioral Economics,” Japan Association for Evolutionary Economics Online Lecture Series,

July 19, 2025.

8. **Ayato Kitadai**, “Comment on ‘Keizai Seminar’ August-September 2024 Issue,” Keizai Seminar December 2024 - January 2025 Issue.
9. **Ayato Kitadai**, “Economic Experiments with AI as Subjects,” Joint Research Project Management Support Workshop (Kyoto Sangyo University), 2024.
10. **Ayato Kitadai**, “Exploring Various Methods for Generative Agents to Replicate Results in Economic Experiments of the Ultimatum Game,” Workshop on Artificial Intelligence in Academic Research, July 9, 2024 (invited).
11. **Ayato Kitadai**, “Decision-making Analysis of Data Provision as a Public Goods Model,” 2nd Mechanism Synthesis Research Seminar, 2024.
12. **Ayato Kitadai**, “Toward a Novel Methodology in Economic Experiments,” Waseda Game Theory and Experimental Economics Workshop, 2024.